

Poetry as a Semantic Landscape

Topographic maps of Poems

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This article explores poetry through the lens of spatial thinking, proposing a new way to interpret poetic texts as landscapes. While metaphors such as “poetic terrain” or “emotional geography” are often used in literary studies, they are rarely developed into structured analytical methods. The aim of this research is to review existing approaches to mapping literature, especially poetry and to introduce a new method that combines cartographic principles with visual interpretation and literary analysis.

Rather than reading poems solely line by line, this approach considers rhythm, structure, and emotional intensity as elements that shape semantic space. In this framework, each poem becomes a landscape shaped by the movement of language and feeling.

The method is based on a simple spatial model: the horizontal axis represents the number of syllables, the vertical axis reflects the line’s position in the poem, and emotional tension is visualized as elevation.

The goal is not just to visualize poetry, but to offer a new perspective for reading it—one that reveals internal structures and affective patterns not always visible through traditional literary analysis. This approach also opens possibilities for applying cartography in the humanities, particularly in visualizing the internal structure and emotional dynamics of poetic texts.

I. MAPPING AS AN EPISTEMIC TOOL

Cartographic and literary studies have long been parallel to each other, but in recent decades they have begun to increasingly closely interact and intersect. The semantic shift which emphasizes from space as place to space as an actor in literature has been described as the “*spatial turn*” (Piatty, 2009) or “*topographical turn*” (Weigel, 2002). In this context, cartography is seen not as a technical “passive practice, but an active process of knowledge production” (Harley, 1989). It allows to explore the semantic, emotional, and symbolic landscapes embedded in poetry.

Furthermore, the mapping of poetic elements highlights the materiality of language and its ability to construct worlds and spaces. As Moretti (1998) writes, “placing a literary phenomenon in its specific space – mapping it – is not the conclusion of geographical work; it’s the beginning. After which begins in fact the most challenging part of the whole enterprise: one looks at the map and thinks”. This means that creating a map is not the final step, but rather the starting point of interpretation that reveals meanings that are not always visible in a traditional reading and offers a new spatial perspective on how poetry works.

II. CONCEPTUALIZING SPACE IN THE LITERATURE ANALYSIS

Literary space is multidimensional “second space” (Moretty, 1998). There are several key concepts that allow abstract meaning to be represented through spatial form: semantic space (Fig.1 a), semantic landscape (Fig.1 b), and imaginary space (Fig.1 c).

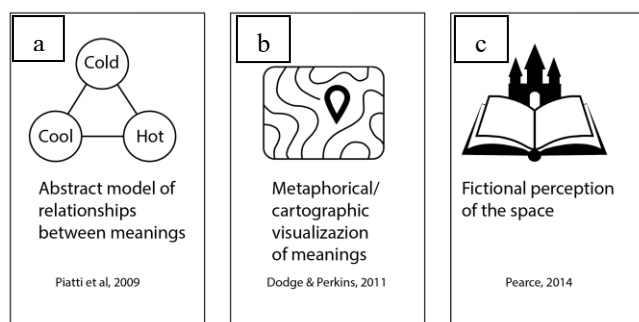


Figure 1. *Spatial forms in Literature*, Piatti (2009), Dodge & Perkins (2011), Pearce (2014)

A. Semantic Space

The concept of semantic space came from linguistics and cognitive science. It refers to an abstract model represents proximity, similarity, or contrast of the meaning and determines spatial relationships between characters and events (Dodge & Perkins, 2011). The theory of Semantic Space includes two main concepts: spatial autocorrelation from geographic information science, which articulate that nearby places tend to share similar characteristics, and the distributional hypothesis from computational semantics, suggesting words with similar meanings tend to cluster together in texts (Gavin et al., 2017).

One example of semantic space is mapping the emotional space time cube introduced by Kalinina, Malikhova, and Kolmogorova (2021) (Fig. 2).

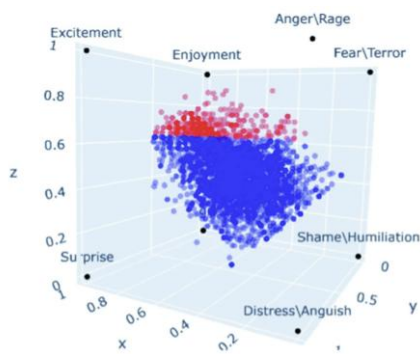


Figure 2. Emotional Space-Time Cube based on the Lövheim Model (Kolmogorova, A., Kalinin, A., Malikova, A. (2021))

This model uses three axes: serotonin, dopamine, and noradrenaline to position texts according to linguistic features associated with specific affective states. This spatial emotional framework represents semantic space which links textual meaning with affective positioning in a three-dimensional cube.

B. Semantic Landscape

Semantic landscape is a spatial representation of emotional and associative meanings in a text. Unlike semantic space, semantic landscape is spatially grounded and can belong to both real and imaginary geographical locations (Piatti et al., 2009; Pearce, 2008). A key example of the semantic landscape in literary cartography is the Holocaust Geographies Collaborative [9]. This project maps the lived experiences of individuals during the Holocaust by visualizing layers of trauma, fear, loss, and memory (Fig. 3).



Figure 3. Emotional Semantic Landscape of Holocaust Narrative [10].

Rather than focusing on physical locations, it reveals how certain places became emotionally and symbolically charged, offering a deeper understanding of how space can carry the weight of personal and collective history.

Describing the meaning of the semantic landscape, Woods writes:

"They describe human lives without ever showing us any people. Instead, we see the underground structures that humans build for waste and the paths they make for squirrels in the sky. We see which homes have wind chimes and which ones call the cops. We see the rout of the letter carrier and the life cycle of the daily paper. Wood is writing a novel where we never meet the main characters, but their stuff is everywhere. I don't know exactly how to describe the feeling that creates. It's like walking around a world that's been decimated by a neutron bomb and walking into all the houses" (Wood, 2010).

Plenty scholars have significantly contributed to the development of the semantic landscape concept through diverse mapping projects (Wood, 2010; Nold, 2009; Bodenhamer et. al, 2015).

A notable example is the Literary Atlas of Europe, developed as part of the European COST Action initiative “Literary Geographies” [11]. This digital platform maps the geographies of European novels, combining real and imagined locations to visualize the literary construction of Europe. Atlas demonstrates how semantics interacts with physical geography, revealing the spatial dynamics of identity, memory, and culture.

A further methodological contribution to the concept of semantic landscape in poetry comes from Cooper and Gregory (2011), who applied literary GIS techniques to poetic representations of the English Lake District. Their work highlights how narrative journeys through space encode emotional and symbolic meanings, transforming geographical landscapes into layered literary terrains.

C. Imaginary Space

Imaginary Spaces include narrative spaces and fictional worlds which «don't have definite borders» (Piatti, 2009). These places are subjective geographies and are closely linked to cultural context and memory. Although the Imaginary space is not real in a physical sense, it “possesses a coherent internal geography that can be spatialized” (Piatti, 2009). One of the illustrative examples of imaginary spaces cartography is the project Mapping the Lands of Middle earth (Fonstad, 1981) (Fig. 4), which focuses on the richly detailed fictional world created by J.R.R. Tolkien. This reconstruction shows how non-geographical spaces can have an internal spatial logic.

Together, the above concepts highlight the different dimensions at which meaning becomes spatial and subject to cartographic representation.



Figure 4. La Carta della Terra di Mezzo Principe, Quirino 1974 [30].

III. POETRY AS A MAPABLE SPACE

Poems construct their own internal geographies, which may be translated into cartographic representations. The spatiality of a poem extends beyond its semantic landscape to include formal elements such as line breaks, typographic arrangements, and rhythmic modulation, all of which contribute to a multidimensional poetic terrain.

A. Phonetic Dimension

Phonetic space refers to the sound patterns within a poem—such as rhyme, assonance, consonance, and alliteration that shape its emotional atmosphere and generate an acoustic sense of spatiality (Delmonte, 2016).

A notable example of phonetic mapping in poetry is the Poemage project, developed by McCurdy, Lane, and Meyer (2015), which visualizes sound patterns as intersecting trajectories across the lines of a poem (Fig. 5). The authors refer to these interwoven structures as “sonic topologies”, allowing the reader to navigate the poem spatially through its acoustic landscape.

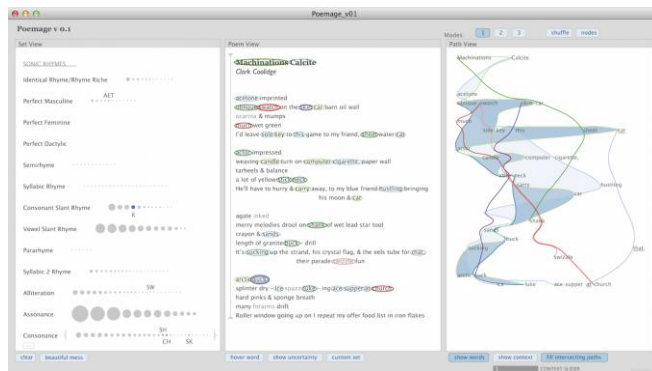


Figure 5. Visualizing Sonic Topologies with Poemage. [19].

In this context poetic text becomes a terrain of affect, association, and rhythm which is open to mapping not geometrically but symbolically” (Cooper, 2011).

B. Metrical Dimension

This dimension is based on rhythm, structure, and the various metrical patterns that form the internal framework of the poem. This structural-metrical space is particularly important within the structuralist approach, since it is what makes the analysis and mapping of the poetic text possible (Horváth, 2019).

Mapping the rhythm of a poetic work incorporates digital poetry analysis methods such as numerical notation systems (e.g., SPARSAR), where scholars assign binary values to syllables—typically “o” for unstressed and “r” for stressed syllables—to create a visual representation of metrical patterns (Delmonte et al., 2014). More sophisticated methods include the analysis of the evolution of poetic complexity, such as the time-series approach introduced by Mansilla, which characterizes the regular sequence of rhythms—patterns of stressed and unstressed syllables in the poem (Mansilla et al., 2002).

A recent advancement in this area is the work by Benito-Santos et al. (2023), who propose a scalable visualization system that displays poetic rhythm as color-coded heatmaps across thousands of poetic lines (Fig. 6)

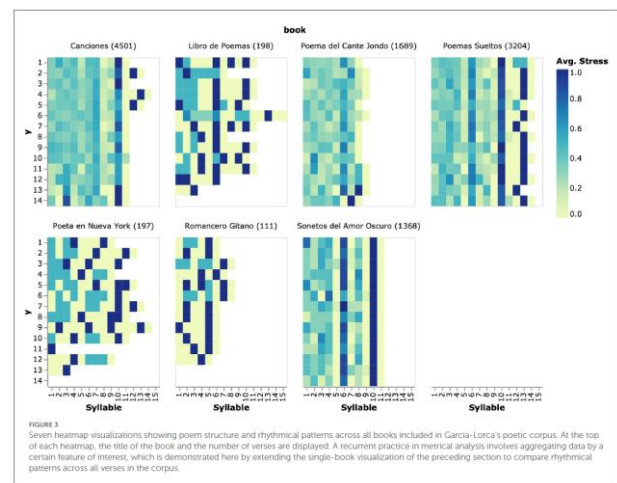


Fig. 6 Heatmap Visualization of Rhythmic Patterns in García Lorca's Poetry [24].

C. Narrative dimension

The narrative dimension of poetry can be mapped as a semiotic network of relationships between events and temporal shifts. Unlike phonetic or metrical structures, narrative cartography is more complex, as poetic narratives are often fragmented, nonlinear, and layered with symbolic meaning.

Several projects have explored narrative mapping in poetry. The Poetry Atlas [25] maps geographic references found in poems. For instance, Wordsworth's “Composed upon Westminster Bridge” allowing readers to experience the poem's physical journey as a form of geographic storytelling. Similarly, the Literary Atlas of Wales [26] visualizes the spatial structure of Welsh literary texts, including poems, by geolocating narrative settings and mapping the movement of characters and events. This approach connects real and imagined landscapes, revealing the spatial logic of poetic narrative.

IV. VISUALISATION OF POETIC TOPOGRAPHY: Mapping Rhythm and Emotion

A. Objective

The aim of this study is to try to combine previous studies and transform poetry through rhythm and meter into a spatial surface and to consider the application of cartographic techniques in the humanities.

B. Methodology

The methodology consisted of three main stages: data preprocessing, metric extraction, and spatial visualization [29].

a. Data processing

For the purposes of visualization, the selection includes two poems with distinct rhythmic and emotional: "The Lake Isle of Innisfree" by W. B. Yeats and "Lady Lazarus" by Sylvia Plath. The poems were segmented into discrete textual units and stored as a one-dimensional Python list, `poem_lines`. This data structure preserved the sequential topology of the text.

b. Metric Extraction

To estimate rhythmic diversity, normalized standard deviation of token lengths per line was implemented (Fig. 7) (Plamondon & Drake, 2010).

This method allows to quantify intra line variation in word length, which is a proxy of the irregularity of the rhythm (Fig. 8)

```
def rhythm_variation(line):
    words = line.split()
    lengths = [len(w) for w in words if w.isalpha()]
    return np.std(lengths) / np.mean(lengths) if len(lengths) > 1 else 0.0

rhythm_score = [rhythm_variation(line) for line in poem_lines]
```

Fig. 7 Python code used for estimating scalar score for each line

```
line = "I have done it again"
words = ["I", "have", "done", "it", "again"]
lengths = [1, 4, 4, 2, 5]
mean = 3.2
std = ~1.47
result = 1.47 / 3.2 ≈ 0.46
```

Fig. 8 Illustration applied method to line "I have done it again"

c. Estimation of the Emotional Intensity

To estimate the emotional intensity of each line, a neural network model based on the transformer architecture was used j-hartmann/emotion-english-distilroberta-base from Hugging Face [28] (Fig. 9).

```
emotion_model = pipeline("text-classification", model="j-hartmann/emotion-english-distilroberta-base", top_k=1)
emotion_intensity = []
for line in poem_lines:
    res = emotion_model(line)[0]
    emotion = res[0]
    score = emotion['score']
    emotion_intensity.append(score)
```

Fig. 9 Neural model based on transformer architecture

d. Spatialization

To transfer the poem to the spatial model, a matrix of heights was designed. The ordinate axis corresponds to the poetic line (`n_lines`), the abscissa axis represents a sample point along that line (`line_length`) defining the horizontal resolution of the grid. The rows were converted to a sinusoidal form obtained from (`rhythm_score`) with additional (`emotion_intensity[i]`). To avoid

homogeneity of the topography, it was decided to add local noise. The final height values per row (`base[i]`) were formed by additional shifts in emotional intensity. Rhythm was represented by structural undulations and emotions by vertical scaling (Fig. 10). To produce a smooth surface, a Gaussian smoothing filter was applied.

```
n_lines = len(poem_lines)
line_length = 60
np.random.seed(0)
base = np.zeros((n_lines, line_length))

for i in range(n_lines):
    wave = np.sin(np.linspace(0, rhythm_score[i] * 10 * np.pi, line_length)) * rhythm_score[i]
    noise = np.random.normal(0, 0.1, line_length)
    height = (wave + noise) * emotion_intensity[i]
    base[i] = height + emotion_intensity[i] * 0.5

elevation = gaussian_filter(base, sigma=1.3)
```

Fig. 10 Generation of elevation matrix

e. Visualisation

For visualization, a topographic map was used Matplotlib (Fig. 11).

```
fig, ax = plt.subplots(figsize=(14, 8))
cmap = plt.get_cmap("terrain")
contour = ax.contourf(elevation, levels=40, cmap=cmap)
lines = ax.contour(elevation, levels=40, colors='black', linewidths=0.3, alpha=0.5)

for i, line in enumerate(poem_lines):
    ax.text(line_length + 1.5, i, line, fontsize=9.5, va='center', ha='left', color='black')

ax.set_xlim(0, line_length + 25)
ax.set_ylim(n_lines, -1)
ax.set_title("Topographic Map of Rhythm and Emotion", fontsize=14)
ax.set_xlabel("Word Position (per line)", fontsize=12)
ax.set_ylabel("Poem Line Number", fontsize=12)

cbar = plt.colorbar(contour, shrink=0.7, pad=0.02)
cbar.set_label("Elevation = Rhythm * Emotion", fontsize=10)
plt.tight_layout()
plt.savefig("poetic_map.svg", format='svg')
```

Fig. 11 Data visualisation

The 3D rendering was done with Three.js using a shader and a vertex mesh. The mesh was built as a parametric surface of a height array.

V. RESULTS AND INTERPRETATION

The results are shown in Figures 12 and 13. A topographic landscape of two different poems has been created. As can be seen from Figure 12, the poem "The Lake Isle of Innisfree" by W. B. Yeats has a smoother relief without sharp, high, small jumps, which indicates a smoother rhythm and less emotional tension.

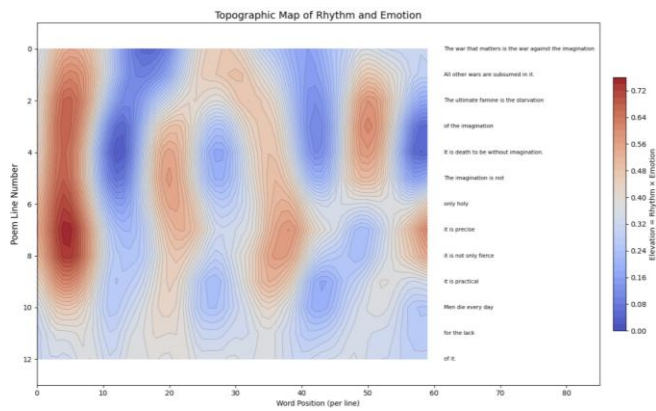


Fig.12 Topographic Map of Rhythm and Emotion “The War Against the Imagination” by Diane Di Prima

The results of the visualization of the topographic relief of the second poem show that it has more “hills” and “elevations”, which indicates a less flat area.

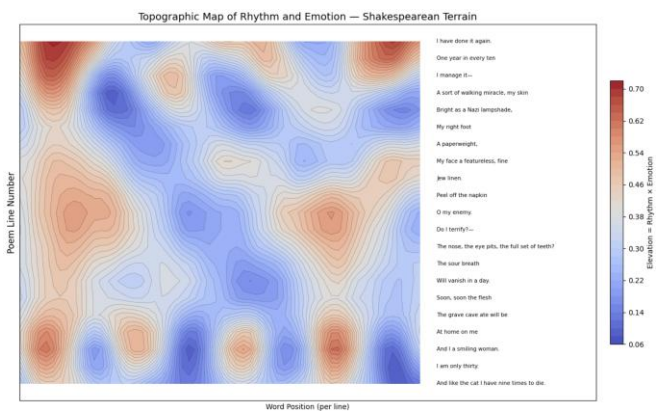


Fig.13 topographic Map of Rhythm and Emotion “Lady Lazarus” by Sylvia Plath

To complement the 2D topographic maps, a three-dimensional visualization was implemented using Three.js, a WebGL Three.js. (Fig. 14, 15).

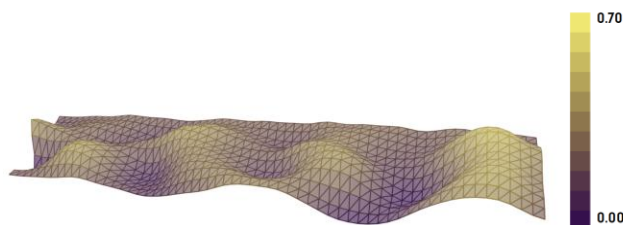


Fig.14 3D Terrain of Poetic Structure – Diane Di Prima

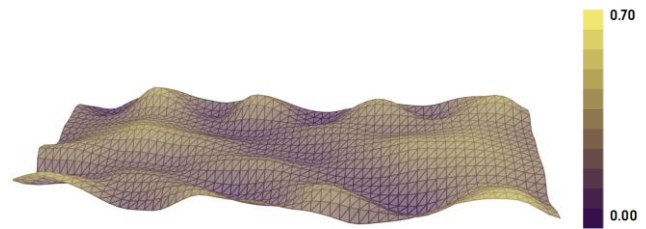


Fig.15 3D Terrain of Poetic Structure – Sylvia Plath

a. Method limitations

1. Emotional intensity could not be evaluated fully as for estimation pre trained model was used.
2. Rhythm was estimated by statistical variables and doesn't count pauses or phonetic prosody.
3. The elevation field resolution and Gaussian smoothing affect surface readability and may distort small-scale features.

VI. CONCLUSION

The study presents the research that has been done in the field of mapping literature, particularly poetry. A new method for creating topographic maps of poetry is proposed, based on statistical variables and visualizing rhythm and meter. The method opens new possibilities for digital literary analysis, data-driven poetics, and poetic visualization.

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